

Temporal Pixel Masking for Mitigating OCR-Based Text Recovery from Short-Exposure Screen Photography: A Feasibility Study with a UVC Camera

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[PLACEHOLDER: Funding acknowledgment to be filled.] This manuscript includes a minimal-risk human-subject usability study. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type. Before each session, the experimenter individually confirmed informed consent and photosensitivity-screening eligibility in person. The study also followed the withdrawal safeguards described in Section V-E, and only de-identified data are analyzed and reported.

ABSTRACT Screen content can be covertly photographed and transcribed by optical character recognition (OCR) or vision-language models (VLMs). We present temporal pixel masking, a display method exploiting differences between human temporal integration and short camera exposures. A base frame is decomposed into rapidly alternating, cryptographically randomized complementary subframes; noise, stripe/glyph overlays, and a partial inversion frame can reduce recoverability. Short rolling-shutter exposures may capture temporally mixed fragments rather than fully integrated frames. We evaluate a 240 Hz software prototype using one USB Video Class (UVC) webcam across nine geometries and 10,575 captures, best-of-engine OCR character recovery decreased from 94.1% unprotected to 15.1% for the readability-priority profile and 5.0% for the high-suppression profile; exact-match rates decreased from 65.7% to 0.4% and 0%, respectively. The latter meets the descriptive $<10\%$ recovery target. Video temporal averaging of 150 frames recovered 47.9–71.1% of characters, and the strongest of three tested VLMs reached 77.8% exact match on 0.5 m short-exposure captures, with recovery concentrated on large-font short snippets. Under long exposure, masking could increase recovery over the unprotected baseline, consistent with reduced luminance shifting the sensor into its linear range. These results establish feasibility for mitigating conventional OCR in the tested short-exposure UVC-camera link, but not protection against temporal integration or VLM attackers. In 56 participants, readability-priority masking retained 94.83% typing accuracy but reduced speed by 1.14 WPM; readability, stability, and comfort averaged 3.79, 2.98, and 3.39 out of 5, indicating measurable usability costs.

INDEX TERMS Adversarial perturbation, camera capture, display privacy, optical character recognition, persistence of vision, screen-camera capture, temporal masking, visual eavesdropping

I. INTRODUCTION

SCREENS routinely display credentials, financial records, source code, and other sensitive information in offices, banks, hospitals, and shared workspaces. At the same time, smartphones, smart glasses [1], and miniature cameras make physical screen capture increasingly unobtrusive. A single photograph can be processed by optical character recognition (OCR) or a vision-language model (VLM) without accessing the host device or leaving a network trace. Controlled experiments and user studies further show that visual eavesdropping

can succeed quickly and often goes unnoticed [2], [3].

Existing defenses address different parts of this threat. Physical privacy filters and spatial display transformations restrict viewing geometry, while digital rights management and watermarking target digital copying or post-capture attribution [4], [5]. Temporal display techniques have also used persistence of vision and psychovisual modulation to separate what viewers and cameras perceive [6]–[10]. These studies establish temporal modulation as a viable display mechanism, but they leave an important empirical question

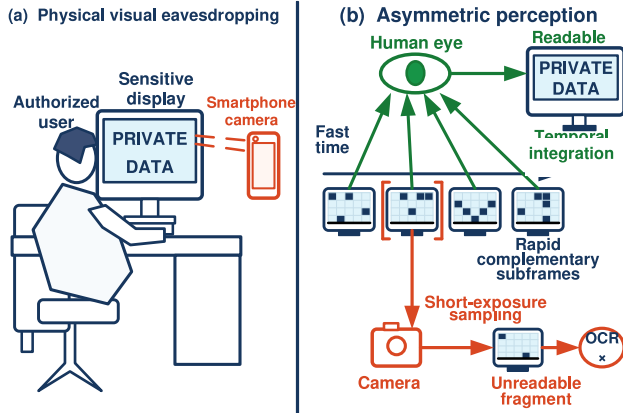


FIGURE 1. Threat scenario and operating principle of temporal pixel masking. (a) A nearby camera can capture sensitive on-screen content without accessing the host system. (b) Human vision integrates complementary subframes, while a short camera exposure can intersect only part of a display cycle and leave OCR with an incomplete fragment. Long exposure or multi-frame reconstruction can accumulate more information.

unresolved. It remains unclear how much sensitive text is machine-recoverable from physical short-exposure captures and how rapidly recovery increases with stronger recognition and temporal reconstruction methods.

We investigate this question through *temporal pixel masking*, a display-side implementation that builds on prior temporal masking principles. Each frame is decomposed into n rapidly alternating subframes whose pixel assignments are randomized, mutually exclusive, and collectively complete in the ideal digital domain. A human observer can integrate information across the display cycle, whereas a short camera exposure may intersect only part of that cycle and record incomplete character structure (Fig. 1). Profile-dependent stripe and glyph overlays, complementary noise, and a partial inversion frame provide additional operating points between readability and suppression. This mechanism does not prevent reconstruction in general: a long exposure or registered multi-frame sequence can accumulate the complementary information.

We implement the method as a 240-Hz graphics processing unit (GPU) rendering prototype and evaluate it with an instrumented USB Video Class (UVC) capture pipeline. The physical archive contains 10,575 captures from one webcam across three distances and three viewing angles. In the primary matched analysis at the common nominal exposure setting, conventional OCR character recovery decreases from 94.5% for the unprotected display to 17.8% for the readability-priority profile and 5.6% for the high-suppression profile. These results provide evidence of feasibility within the evaluated short-exposure camera link rather than a general guarantee against screen photography.

The evaluation therefore extends beyond favorable single-frame conditions. Fixed image preprocessing, long exposure, 150-frame temporal averaging, digital reconstruction, and VLM probes test how recovery changes as attacker capa-

bility increases. These attacks recover substantial information, especially from large-font short snippets, and expose the integration boundary of the approach. A 56-participant study separately measures the human side of the trade-off: readability-priority masking retains high typing accuracy but imposes measurable costs in speed, perceived stability, and visual comfort.

The main contributions are as follows:

- 1) **System-level physical evidence:** We provide a controlled measurement of short-exposure screen capture using archived physical images, multiple content types and geometries, three OCR engines, explicit sensitive-field recovery, matched profile comparisons, and content-cluster resampling. This evaluation quantifies temporal pixel masking as a system behavior rather than inferring protection from digitally generated subframes alone.
- 2) **Attack-boundary characterization:** We evaluate fixed preprocessing, long-exposure capture, temporal-average attacks, digital reconstruction, and commercially hosted VLMs. The resulting evidence identifies both the conditions under which conventional OCR recovery decreases and the conditions under which sustained capture or stronger recognition recovers substantial content. Detection and tracking experiments provide secondary cross-task stress tests.
- 3) **Security-usability trade-off:** We compare readability-priority and high-suppression operating points and conduct a controlled within-subject study with 56 participants. The results connect machine recovery to typing performance, readability, perceived stability, and immediate visual comfort, making the cost of the usability-facing configuration explicit.

The remainder of this paper reviews related work, defines the threat model and system design, and then presents the physical, attack-oriented, and usability evaluations. The final sections discuss the evidence boundaries and conclude the study.

II. RELATED WORK

A. SCREEN VISUAL EAVESDROPPING THREATS

Sensitive screen content can be acquired through direct photography, optical reflections, or non-visual side channels. Backes et al. [11] recovered displayed content from reflections on nearby objects, while Raguram et al. [12] reconstructed typed input from reflections and refractions in a user's surroundings. Electromagnetic attacks remove the need for an optical line of sight: Screen Gleaning recovered security codes from mobile-display emissions [13], and Deep-TEMPEST used learned reconstruction to recover text from unintended HDMI emissions [14]. Camera glasses further increase the likelihood of inconspicuous recording, although recent work by Wang et al. [1] studies wearer-bystander privacy expectations and protection preferences rather than the technical recovery of screen content.

Human-centered studies establish the practical relevance of these threats but do not by themselves characterize machine-readable leakage. An industry-sponsored visual-hacking experiment reported successful information capture in 91% of trials, with the first success occurring within 15 minutes in 49% of trials [2]. Eiband et al. [3] collected 174 accounts of shoulder-surfing incidents and found that observation was commonly opportunistic and often went unnoticed. These studies motivate display privacy, whereas the present work focuses on a narrower technical question: how much static text remains recoverable from short-exposure physical captures, and how that recovery changes under stronger acquisition and post-processing attacks.

B. DISPLAY PRIVACY PROTECTION

Existing defenses intervene at the camera, the display, or the surrounding illumination. ScreenAvoider detects displays and sensitive applications in images so that a cooperating capture or disclosure pipeline can suppress them [15]; its protection therefore depends on control of the camera or downstream image handling. Display-side spatial methods instead alter what an observer sees. HideScreen suppresses low-frequency visual cues so that content blends into the background beyond a designed viewing distance [4]. Eye-Shield re-renders content to preserve close-range readability while reducing recognition at larger distances and viewing angles [16]; in addition to user studies, it evaluated recognition on photographs using cloud OCR. Both systems primarily address distance- and angle-dependent shoulder surfing rather than temporally sampled display content. Illumination-based systems such as LiShield [17] and Rainbow [18] exploit rolling-shutter capture by modulating ambient light, but they protect illuminated physical scenes rather than altering the displayed pixels themselves.

Attribution mechanisms address a different stage of the leakage process. General image watermarking embeds identifiers that may survive subsequent processing [19], and screen-shooting-resilient schemes explicitly model the display-camera channel [5]. The mID system similarly creates camera-visible moiré identifiers for tracing leaked screen photographs [20]. These approaches can identify a leakage source after capture, but they are not designed to prevent an attacker from reading the photographed content. Visual cryptography is also conceptually related because it decomposes an image into shares that reveal the secret only when combined [21]; its original threat model, however, concerns cryptographic share construction rather than time-varying display capture.

Temporal display techniques are the closest predecessors of the present work. Yerazunis and Carbone introduced time-masked privacy-enhanced displays [6]. Wu and Zhai subsequently formalized temporal psychovisual modulation [8], and Gao et al. applied that framework to an information-security display [9]. Lim used persistence of vision to construct browser keypads that resist a single digital screenshot [7]. Kaleido decomposes video into chrominance-

complementary frames with controlled luminance variation so that viewers perceive the intended sequence while camera recordings are degraded [10]. Screen-camera communication explores the complementary objective: systems such as spatially adaptive embedding, DeepLight, and passive screen-to-camera communication encode camera-decodable information while limiting perceptual disturbance to human viewers [22]–[24].

These studies establish temporal modulation and multi-frame decomposition as prior mechanisms; the present work does not claim to introduce either concept. The unresolved issue is their security boundary for static sensitive text under modern machine recognition and increasingly capable capture strategies. Existing studies do not jointly measure short-exposure physical capture, multi-engine OCR, vision-language-model recovery, temporal integration, and phase-aware reconstruction. Our implementation also differs from Kaleido in mechanism: it uses randomized discrete pixel partitioning, and its exploratory high-suppression profile deliberately breaks strict per-cycle completeness (§IV-E). Table 1 compares representative systems by protection locus, threat model, and reported evaluation.

C. MACHINE RECOGNITION AND PHYSICAL-WORLD ROBUSTNESS

Research on adversarial examples provides a second point of comparison because it separates human readability from machine recognition. Song and Shmatikov showed that small digital modifications to text images can induce targeted OCR errors while preserving the apparent meaning for human readers [25]. More general digital attacks optimize model-specific perturbations [26]–[28], whereas physical attacks must remain effective after changes in viewpoint, illumination, printing, deformation, and camera processing [29]–[31]. Athalye et al. introduced Expectation over Transformation (EoT) to optimize adversarial examples over a chosen transformation distribution and demonstrated robust physical 3D objects [32]. Surveys of physical attacks and adversarial transferability show that robustness across capture conditions and generalization across models are distinct empirical questions [33], [34].

The display-camera channel adds transformations that are absent from ordinary digital evaluation. Niu et al. showed that simulated moiré patterns from screen photography can act as transferable adversarial perturbations [35]. Conversely, learned demoiréing can recover screen photographs degraded by such artifacts [36], illustrating why incidental optical distortion should not be treated as a stable security guarantee. The base mechanism studied here does not optimize a spatial perturbation against a particular recognizer; it partitions the displayed content over time. Enhanced profiles add noise and stripe or glyph overlays, but their effectiveness is measured after physical capture across multiple OCR and vision-language models rather than inferred from digital attack success. This distinction motivates our system-level evaluation of recovery and attack boundaries.

TABLE 1. Scope comparison with representative screen-privacy systems. Entries describe different threat models and evaluation targets; they are not matched-parameter performance claims.

Scheme	Protection locus	Target threat	Mechanism	Reported evidence and boundary
ScreenAavoider [15]	Camera or disclosure pipeline	Incidental capture of sensitive screens	Screen and application recognition; policy-based suppression	Lifelogging-image detection; requires a cooperating capture or disclosure path
HideScreen [4]	Mobile display	Human viewing beyond a designed distance	Spatial suppression of low-frequency cues	Human text/image recognition and usability across viewing distances
Eye-Shield [16]	Mobile display	Human viewing at larger distances and angles	Distance- and angle-aware spatial re-rendering	Human studies and photographed-content recognition; spatial rather than temporal defense
LiShield and Rainbow [17], [18]	Ambient illumination	Rolling-shutter photography of physical scenes	Temporally modulated LED illumination	Camera-image interference; requires controllable illumination and rolling-shutter interaction
Yerazunis and Gao [6], [9]	Displayed image	Incomplete temporal observation	Time masking or temporal psychovisual modulation	Proof-of-concept security displays; no modern OCR/VLM attack suite
Kaleido [10]	Displayed video	Continuous mobile-camera recording	Chrominance-complementary frames and luminance variation	PSNR, structural similarity, and subjective quality for protected video
This work	Static screen text	Short-exposure capture followed by machine recognition	Randomized pixel partitioning; high-suppression profile breaks completeness	OCR/VLM recovery, temporal integration, phase-aware reconstruction, and usability

III. THREAT MODEL

A. PROTECTED ASSETS AND EVALUATED CONTENT

The motivating assets include passwords, financial data, personally identifiable information (PII), source code, and confidential documents. The corpus represents these assets through credentials, digit strings, code, paragraphs, mixed-language snippets containing embedded URL paths, and a document page. Eighteen high-value fields in 9 of 12 items were annotated before scoring; ordinary prose and incidental section numbers were excluded.

B. ATTACKER CAPABILITIES

Attack capabilities do not form a strict monotonic hierarchy but are jointly determined by exposure, geometry, temporal sampling, and post-processing. Results are reported across the following scope:

- **Primary scope:** Instrumented snapshots at the nominal 3.91 ms UVC exposure-control setting, followed by per-frame recognition with Surya, EasyOCR, and Tesseract. The primary result uses raw camera crops. A separate three-engine stress test selects per capture among five predeclared deterministic transforms plus the raw input.
- **Extended attacks:** Long exposure, 150-frame temporal averaging, off-axis capture, phase-aware channel reconstruction, object detection and association, and commercially hosted vision-language models (§V-D).

C. ATTACKER KNOWLEDGE

The attacker *knows* the temporal masking algorithm, the sub-frame count n , and candidate period parameters, but does *not know* the current cycle's mask seed or phase starting point. The cryptographically secure pseudorandom number generator (CSPRNG) provides per-cycle mask randomization so that no fixed pixel-to-slot pattern repeats across cycles; it does not provide a cryptographic guarantee against full-cycle

averaging attacks. Once an attacker obtains and registers a complete cycle, an unknown seed cannot prevent linear reconstruction. Sparse sampling may disrupt glyph evidence used by conventional OCR, but the present design does not isolate binarization, segmentation, or recognition as the responsible stage. End-to-end VLMs recovered appreciable text from the same archived fragments (§V-D), so the conventional-OCR observation does not transfer to those models.

D. PROTECTION GOALS

We use tiered design goals under short exposure to interpret the evaluated profiles. High-suppression targets conventional OCR character recovery $< 10\%$ with exact match $= 0\%$, whereas readability-priority targets a relative reduction of at least 80% with exact match $< 1\%$. These descriptive thresholds organize the profile comparison. Object-detection mean average precision at 0.50 intersection over union (IoU) threshold (mAP50) provides a secondary diagnostic, while ΔE_{00} and the structural similarity index (SSIM) quantify digital reconstruction quality.

IV. SYSTEM DESIGN

A. TEMPORAL SUBFRAME DECOMPOSITION

Given a normalized original frame $\mathbf{I} \in [0, 1]^{H \times W \times C}$, we generate binary masks $\mathbf{M}_1, \dots, \mathbf{M}_n$ satisfying $\sum_{k=1}^n \mathbf{M}_k = \mathbf{1}$, and set $\mathbf{I}_k = \mathbf{I} \odot \mathbf{M}_k$. This yields:

- **Digital-domain completeness:** $\sum_{k=1}^n \mathbf{I}_k = \mathbf{I}$;
- **Mutual exclusivity:** each pixel is illuminated in exactly one of the n time slots.

The distinction between “summation” and “temporal averaging” matters: on an ideal linear display with equal slot durations and equal peak luminance, the average radiance over one cycle is $\mathbf{I}/n$, not $\mathbf{I}$, so digital-domain completeness means only that subframes can be re-summed, not that real display luminance has been losslessly preserved.

Pixel assignments are generated from a ChaCha20 [37] CSPRNG, with rejection sampling mapping the byte stream to unbiased slot indices and a separate Fisher–Yates shuffle [38] randomizing subframe playback order. A chi-square slot-count check ($\alpha = 0.01$, $df = n-1$; 3 at the evaluated $n = 4$), used only as an implementation sanity check with at most 5 resampling attempts, is not evidence of cryptographic strength. Per-cycle randomization avoids a fixed, periodically repeated pixel-to-slot pattern across cycles, but it cannot prevent linear reconstruction of a registered full cycle. This ChaCha20-based generator drives the Python physical-capture pipeline. The user-study web implementation reproduces the same mask structure, rejection sampling, and playback shuffling with a deterministic seeded 32-bit generator so that stimuli are exactly repeatable in the browser; it has no CSPRNG properties (§V-E). Fig. 2 illustrates the complete pipeline.

B. COMPLEMENTARY ADVERSARIAL NOISE

We overlay adversarial noise $\mathbf{N}_1, \dots, \mathbf{N}_n$ on the subframes, satisfying:

$$\sum_{k=1}^n \mathbf{N}_k = \mathbf{0} \quad (1)$$

Equation (1) holds strictly only in the ideal linear domain prior to summation, clipping, and the display transfer function. Motivated by prior work showing that small text-image perturbations can significantly mislead OCR systems [25], noise is generated per cycle by single-step fast gradient sign method (FGSM) [26] or iterative sign projected gradient descent (PGD) [27] updates under an $\epsilon = 8/255$ bound, rotating across eight OCR and detector targets; depending on availability, gradients come from the EasyOCR recognition network, a differentiable shadow objective, or a deterministic image-space surrogate. The resulting noise is then decomposed complementarily in the temporal domain. The user-study web implementation substitutes a deterministic image-space texture of comparable amplitude for these model-gradient perturbations (§V-E). Because displays cannot emit negative radiance, the implementation uses a pedestal level and numerical clipping; consequently, after gamma correction, quantization, clipping, and camera imaging, the noise is not guaranteed to cancel physically.

Component role: The random dot-pattern mask is the primary temporal mechanism, while complementary noise is an optional ideal-domain component and stripe/glyph overlays together with partial inversion are nonlinear additions used by the enhanced profiles below. The archived composite-profile definitions preserve correspondence with the captured data, and the physical analysis compares both individual ablations and complete profiles.

C. LUMINANCE MODEL ACROSS RENDERING PATHS

Because each pixel is illuminated only once per n -slot cycle, restoring the original cycle-averaged luminance in a linear-light model would require a factor- n light-output increase

during the active slot. No evaluated path implements per-slot backlight control. The physical-capture playback fixes the compositor gain at $\gamma = 1$ under fixed panel settings, leaving the linear-model cycle average at $\mathbf{I}/n$. The user-study web implementation instead applies a pixel-space gain of $1.1n$ with clipping to $[0, 255]$, which partially compensates the $1/n$ duty cycle but clips bright content; its stimuli are therefore not photometrically identical to the physical-capture condition. Digitally integrated reconstructions are normalized before comparison, and CIEDE2000 ΔE_{00} [39] and SSIM [40] quantify these digital reconstructions only. None of these settings establishes radiometric equivalence on a real panel.

D. TIMING AND BANDWIDTH

For a basic cycle containing only n complementary subframes, taking 60 Hz as the minimum cycle rate requires $f_r \geq 60n$; at $n = 4$ this corresponds to 240 Hz. Including one inversion frame changes the requirement to $f_r \geq 60(n+1)$, dropping the actual cycle rate to only 48 Hz at 240 Hz. Thus, while 240 Hz meets the basic four-subframe configuration, it falls short of the inversion-frame configuration's 60 Hz cycle target. Because human flicker perception also depends on luminance, contrast, retinal eccentricity, spatial content, and eye movements [41], [42], nominal cycle rate alone cannot predict comfort.

Whether an observer integrates complementary subframes into a perceptually complete image depends on the temporal contrast sensitivity function (TCSF) rather than on a single critical flicker fusion (CFF) cutoff; with high-frequency spatial edges, flicker can remain visible above 500 Hz [42]. The 48 Hz full-cycle rate of the inversion configuration may therefore produce visible flicker for some observers, the 60 Hz basic configuration cannot be assumed flicker-free, and the user study in §V-E evaluates readability and immediate comfort under these configurations.

E. EXPLORATORY HIGH-SUPPRESSION PROFILE

Because the basic linear temporal mask leaks under full-cycle temporal averaging (see §V-G2), we instantiate composite operating profiles that add nonlinear components to probe the suppression–utility trade-off and the system's recovery boundaries. Unless otherwise stated, profiles in this subsection use $n = 4$ base temporal masking with complementary adversarial noise enabled. Table 2 summarizes all profile configurations; the “Mask + noise” row in Table 5 is the base condition without stripe/glyph anti-OCR overlay.

- **Strong (anti-OCR) profile:** base mask+noise with mask cell size of 1 pixel, stripe width of 10 pixels, stripe amplitude $\alpha_s = 0.10$, and glyph amplitude $\alpha_g = 0.12$, without an inversion frame.
- **Readability-priority profile:** the Strong profile plus an $\alpha = 0.2$ inversion frame, designated in this paper as the usability-facing composite configuration. Archived result tables and capture metadata retain the legacy label “deployed.”

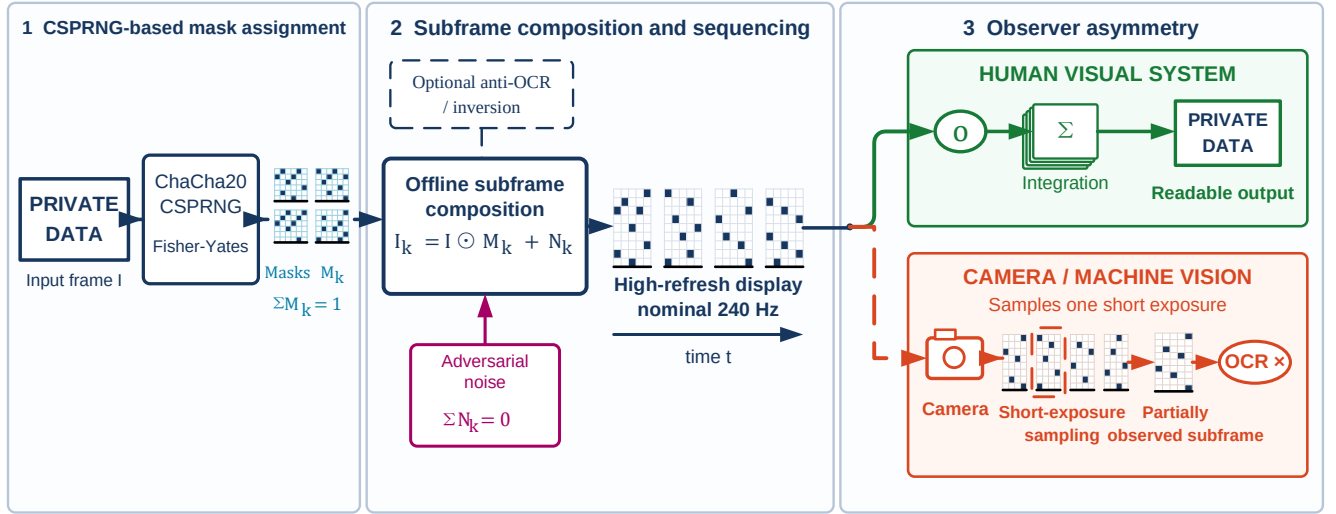


FIGURE 2. Temporal pixel masking system pipeline from source content to protected display. The input frame is decomposed into CSPRNG-assigned complementary pixel masks; optional complementary OCR-oriented noise, profile-dependent stripe/glyph overlays, and an optional partial inversion frame are then inserted before high-refresh playback. A human observer is expected to integrate multiple slots into a readable view, whereas a short-exposure camera may capture only a sparse subframe. Long exposure and video temporal averaging can accumulate multiple slots and are evaluated separately.

TABLE 2. Profile Composition Summary ($n = 4$ for All Profiles)

Profile	Mask	Noise	Cell (px)	Stripe width (px)	Stripe α_s	Glyph α_g	Inv. α
Mask only	✓	–	1	–	–	–	–
Mask + noise	✓	✓	1	–	–	–	–
Strong	✓	✓	1	10	0.10	0.12	–
Readability-priority	✓	✓	1	10	0.10	0.12	0.2
High-suppression	✓	✓	2	6	0.42	0.55	0.2

- **High-suppression profile:** an exploratory security-oriented stress profile using base mask+noise with mask cell size increased to 2 pixels, stripe width of 6 pixels, stripe and glyph amplitudes of 0.42 and 0.55 respectively, plus an $\alpha = 0.2$ inversion frame.

The nonlinear profiles depart from strict completeness ($\sum I_k \neq I$), producing visible stripes and grain. In the static-content capture threat addressed here, completeness is the entry point for integration attacks. The high-suppression profile is designed to trade visible artifacts for reduced full-cycle recovery; §V-B reports the measured outcome.

The paper-facing name “high-suppression” reflects the profile’s relative recovery in Table 3. The archived identifier remains *capture_hardened* (with legacy tag *v1m*) for reproducibility; §V-B reports residual leakage and behavior under the preprocessing oracle.

F. PARTIAL INVERSION FRAME

A long-exposure attack occurs when the exposure window covers a full display cycle ($n = 4$ at 240 Hz yields a basic cycle of approximately 16.7 ms; with an inversion frame the $n+1$ slot cycle is approximately 20.8 ms), where the archived long-exposure labels of 31.25 and 125 ms are nominal values derived from UVC controls rather than photometric measure-

ments. To perturb, rather than eliminate, full-cycle accumulation, the evaluated offline-playback and web pipelines append a **partial inversion frame** $\alpha(1 - I)$ (normalized domain $I \in [0, 1]$; 8-bit form $\alpha(255 - I_8)$) after each n -subframe cycle, implemented as amplitude scaling over one additional display slot because vsync playback holds slot duration fixed. Higher α changes both the integrated signal and the digital reconstruction’s luminance and contrast.

A real-capture ablation across $\alpha \in \{0, 0.2, 0.3, 0.5, 1.0\}$ (§V-C) informs this choice; the readability-priority setting of $\alpha = 0.2$ is a modeled design decision, not a validated optimum.

Each display cycle thus contains $n + 1$ time slots; the resulting 48 Hz cycle rate at $n = 4$ and 240 Hz, and its flicker implications, are discussed in the timing analysis above.

G. IMPLEMENTATION SCOPE AND REPRODUCIBILITY

The proof of concept (PoC) operates at the application layer in Python (NumPy, PyCryptodome, pygame/SDL, mss, ModernGL; PDF input via pypdfium2 or pdftoppm) and separates three rendering paths. The evaluated physical-capture path pre-generates all subframes offline with a CPU compositor and plays the sequence back through pygame with vsync on the 240 Hz panel, so no per-slot synthesis occurs dur-

ing display. A separate live-demonstration window captures the screen with mss and renders through an optional OpenGL backend that reads each frame back to the CPU and falls back to a NumPy renderer when GPU initialization fails; one capture-mask-compose cycle takes on the order of 190 ms, so this window cannot sustain the 120–240 Hz slot budget and serves demonstration only. Its inversion variant shows a full inversion frame for a target fraction α of the slot interval, a duration-based approximation that vsync does not guarantee. The user study runs a third, independent JavaScript/Canvas2D implementation (§V-E). Mask keys default to 32 bytes from the operating-system CSPRNG, and a fixed key can be supplied for regeneration; the playback metadata records permutations, the per-cycle noise schedule, and profile parameters, but not the key, so reproduction relies on the archived stimuli and explicit configurations rather than on command-level replay. No operating-system, display-driver, or backlight-level integration is implemented.

V. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL SETUP

Hardware platform: Most experiments used a desktop workstation with an Intel Core i7-8700K (3.70 GHz), 32 GB RAM, and an NVIDIA TITAN Xp (12 GB VRAM). The display was an AOC 24G51Z driven at a nominal 240 Hz. Each nominal 240-Hz slot lasts 4.17 ms, while the basic and inversion-augmented cycles are 60 and 48 Hz.

Capture device: All physical captures used one eMeet SmartCam S600 USB webcam with its standard UVC driver. Camera frames underwent perspective correction and content cropping before recognition. Per-capture metadata store exposure labels derived from DirectShow/UVC log₂ controls: -11 and -8 map nominally to 0.49 and 3.91 ms, while -5 and -3 map to 31.25 and 125 ms. Eight geometries used the common $-8/-5$ settings; d0.5/15° used $-11/-3$. These values are control-derived labels. The display-brightness setting remained fixed, and video was captured nominally at 60 fps. The primary OCR inputs are geometrically corrected content crops.

The camera was placed at 0.5 m, 1 m, and 1.5 m from the display, rotated to 0°, 15°, and 30° at each distance, providing nine distance-angle configurations for the physical evaluation. The manufacturer specifies an 8-megapixel 4K sensor and 1080p/60 fps output [43].

Test content: Each position used 12 fixed items: two account strings, two code snippets, two digit strings, one English sentence, two mixed-language snippets, two English paragraphs, and one CET6 (College English Test Band 6) document. The CET6 page and two mixed-language items include Chinese characters. Three repeats give 36 captures per profile-attack-position cell and 324 across 9 positions before disclosed repeats. Detection and tracking use separate subsets as cross-task stress tests.

Evaluation metrics: OCR character recovery is defined as $\max(0, 1 - \text{CER})$, where CER is character error rate. Exact match uses case-insensitive full-text comparison after

whitespace removal. Explicit-field recovery uses 18 fields fixed before scoring across 9 field-bearing content items. A field counts as recovered only when its complete Unicode-normalized form appears in at least one engine output for that capture; case, whitespace, and punctuation are ignored. We report field-level micro recovery and sample-level macro recovery separately. Leak rate is the proportion of samples with character recovery $\geq 20\%$ and is only a descriptive threshold.

OCR aggregation and uncertainty: Character and whole-text metrics are maximized independently across the three engines for each capture. Explicit fields use the union of fields recovered by the three engines. Duplicate captures with the same profile, content, position, and repeat are averaged before matching. The primary comparison uses keys present in all three profiles. Paired contrasts resample the 12 content items as clusters with 2,000 resamples and seed 20260612. These intervals quantify sensitivity across the archived content clusters while holding geometry fixed.

Common-setting analysis: The d0.5/15° position used a different nominal UVC setting and produced visibly underexposed protected frames. The remaining 8 geometries define the primary fixed-setting result (Table 3). The unbalanced 9-geometry archive is retained as sensitivity evidence in Table 5.

Fixed preprocessing stress test: We evaluated Tesseract, EasyOCR, and Surya on all 984 selected primary-pool images. The fixed input grid comprised raw, gamma 0.5, luminance contrast-limited adaptive histogram equalization (CLAHE; 1st–99th percentile stretch, clip limit 4.0, 8×8 tiles), unsharp masking ($\sigma = 1.0$, weights 1.7/–0.7), Gaussian adaptive thresholding (block 31, constant 5), and $2 \times$ bicubic upscaling. Parameters were fixed before scoring. For each metric and capture, the attacker retained the best engine-input pair; duplicate captures were then averaged before the same three-profile matching used by the primary analysis. The complete matrix contains 17,712 cells ($984 \text{ images} \times 6 \text{ input forms} \times 3 \text{ engines}$) with no OCR errors. Nonraw Tesseract cells used Tesseract 5.4.1 through `pytesseract 0.3.13`; EasyOCR 1.7.2 and Surya 0.14.7 transformed cells used CUDA on the workstation above. The resulting oracle measures a reproducible fixed-grid preprocessing attack.

Digital reconstruction proxies: ΔE_{00} is the mean CIEDE2000 color difference between the original image and the full-cycle digitally reconstructed image; SSIM [40] is their mean structural similarity. Both metrics support within-paper comparison of digitally reconstructed profiles.

B. REAL-CAPTURE OCR EXPERIMENT

The physical archive comprises 10,575 images (1,175 per position), including ablations, parameter sweeps, and main attack conditions. Readability-priority includes a second capture round for 5 of 12 items, giving 459/153 all-position short/video images rather than 324/108. Duplicate keys are averaged before matched contrasts. Surya [44], EasyOCR [45], and Tesseract [46] produced 31,725 engine-level records. The Surya rerun used `surya-ocr 0.14.7`; the

primary pipeline also used Tesseract 5.4.1 (pytesseract 0.3.13) and EasyOCR 1.7.2. Each geometry uses an archived four-point homography and output canvas, followed by the recorded content crop. The primary OCR pipeline adds no adaptive enhancement. Profiles were collected in the order original, mask-only, mask+noise, Strong, readability-priority, and high-suppression. The main comparison uses 8 common-setting geometries, while Table 5 retains all 9 as sensitivity evidence. Fig. 3 shows digital reconstructions and matched captures at 1.0 m/0°.

Key findings:

- 1) **Primary fixed-setting evidence:** Table 3 is the main short-exposure estimate. On the same 288 matched units per profile (12 content items $\times$ 8 geometries $\times$ 3 repeats), mean recovery is 94.5% unprotected, 17.8% for readability-priority, and 5.6% for high-suppression. The paired content-cluster contrasts are 76.7 percentage points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]). The full 9-geometry captured-image pool gives 94.1%, 15.1%, and 5.0% across the archived settings and repeats. Table 5 gives capture-resampling summaries for the unbalanced archive; across all geometries, unprotected recovery exceeds readability-priority by 78.0 percentage points ([75.5, 80.3]) and high-suppression by 89.1 points ([85.0, 92.4]), while mask-only exceeds readability-priority by 3.0 points ([−0.5, 6.5]). Excluding the noncomparable d0.5/15° geometry raises readability-priority 150-frame video-mean recovery from 71.1% to 79.7% and high-suppression from 47.9% to 53.9%.
- 2) **Fixed preprocessing defines a stronger OCR attack:** On the same 288 matched units per profile, the raw three-engine oracle yields 94.5% unprotected, 17.8% readability-priority, and 5.6% high-suppression recovery. Allowing the five fixed transforms raises these values to 95.9%, 40.2%, and 13.7%, respectively (Table 7). The paired unprotected-minus-readability contrast remains 55.7 percentage points ([50.3, 60.5]), while the unprotected-minus-high-suppression contrast remains 82.2 points ([78.5, 85.2]). The complete 17,712-cell matrix has no OCR errors.
- 3) **Profiles show distinct component and target behavior:** The mask-only condition reaches 19.2% in the pooled archive, while mask+noise reaches 25.6% and Strong reaches 20.7%, revealing nonmonotonic component effects. In the matched pool, readability-priority retains 17.7% field-micro exact recovery (Table 4) with P50/P75/P90/P95/P99 character recovery of 10.1/22.0/45.6/63.6/95.0% (high-suppression: 3.4/9.2/15.2/19.1/22.6%). High-suppression reaches a 5.6% matched mean at the common setting, meeting both its descriptive $< 10\%$ character target and the exact-match = 0% target: no short-exposure capture produced an exact match (Table 5). For readability-priority, the raw common-setting reduction is (94.5 –

17.8)/94.5 = 81.2% with 0.4% exact match, meeting the $\geq 80\%$ relative reduction and $< 1\%$ exact-match targets under the primary analysis. The preprocessing oracle reduces the gap to (95.9 – 40.2)/95.9 = 58.1%, falling short of the same threshold.

- 4) **Integration attacks recover substantial text** (Fig. 4): The readability-priority profile reaches 60.9% char / 36.4% exact under nominal long exposure and 71.1% char / 42.5% exact under the 150-frame mean. Excluding d0.5/15° raises video recovery to 79.7%. Weak inversion ($\alpha = 0.2$) produces little integration reduction, and mask-only and Strong are similar under the 150-frame mean (79.0% versus 78.6%). High-suppression shows lower integration recovery, although substantial text remains recoverable.
- 5) **Long exposure produces a position-dependent reversal:** The unprotected long-exposure baseline is 47.3% char, below the readability-priority profile's 60.9%. At the sole 125 ms position (0.5 m/15°), unprotected recovery is 1.1% versus 46.3% readability-priority, with visible clipping in the full-brightness image. At 31.25 ms, the expected direction holds at 0.5 m, whereas five of six 1.0–1.5 m positions show higher readability-priority recovery. Residual stripe/glyph edges interacting with geometry-specific saturation or contrast compression provide a candidate explanation. Short-exposure recovery remains lower for readability-priority at every position.
- 6) **High-suppression reduces integration leakage:** This exploratory stress profile yields 9.3% char [7.9, 10.9] / 0% exact for nominal long exposure and 47.9% char [41.8, 54.0] / 5.6% exact [1.9, 10.2] under the 150-frame mean. Brackets are capture-resampling summaries. The pooled mean includes considerable interposition variation and a 0% underexposed session. Excluding that session raises video recovery to 53.9% [48.0, 59.9] and exact match to 6.3%. The user study evaluates readability-priority rather than this visibly stronger profile.
- 7) **Geometric trends:** Across 9 conditions (0.5–1.5 m $\times$ 0–30°), protected short-exposure recovery remains below unprotected recovery. Geometry-specific values are reported descriptively because the archived design is unbalanced. A leave-one-round-out analysis on the five repeated content items gives readability-priority short-exposure recovery of 16.6% for round 1 and 15.8% for round 2, leaving the profile-level pattern unchanged.

Per-engine results: Table 6 breaks down pooled raw-input short-exposure results by engine. The engines differ widely in unprotected recovery (Surya 37.1%, EasyOCR 79.5%, Tesseract 84.3%). Under Strong, recovery ranges from 2.1–16.3%; under readability-priority, 3.2–11.7%; under high-suppression, 1.8–2.0%. Table 7 and Fig. 5 use the matched common-setting pool and show that every engine benefits

TABLE 3. Primary Matched Short-Exposure OCR Comparison at the Common Nominal 3.91-ms UVC Setting. Duplicate Captures Are Averaged Before Matching; $N = 288$ per Profile Comprises 12 Content Items $\times$ 8 Geometries $\times$ 3 Repeats. The Noncomparable d0.5/15° Run Is Excluded.

Profile	N	Mean char recovery	Unprotected — profile	Interpretation
Original (unprotected)	288	94.5%	—	Common-setting baseline
Readability-priority	288	17.8%	76.7 pp [74.3, 79.2]	Conventional-OCR reduction, with upper-tail failures
High-suppression	288	5.6%	88.9 pp [85.5, 91.9]	Exploratory stress profile; meets descriptive < 10% target

TABLE 4. Explicit Sensitive-Field Exact Recovery in the Primary Matched Short-Exposure Pool. The 18 Fields Were Fixed Before Scoring; 216 of 288 Units per Profile Contain at Least One Field.

Profile	Field micro (432 opp.)	Sample macro
Original (unprotected)	91.4%	86.8%
Readability-priority	17.7%	17.5%
High-suppression	1.2%	1.1%

from at least one fixed transform. For readability-priority, recovery rises from 3.2% to 10.8% for Tesseract, 14.2% to 33.9% for EasyOCR, and 4.1% to 20.2% for Surya. For high-suppression, the corresponding changes are 2.0% to 8.2%, 2.3% to 6.5%, and 2.0% to 6.0%.

C. INVERSION-STRENGTH ABLATION

A real-capture inversion-strength ablation (Table 8; pipeline per §V-B) found 68.6% recovery at $\alpha = 0$ and 67.0% at $\alpha = 0.2$. Point estimates across $\alpha \in [0, 0.5]$ ranged from 61.7% to 72.0%; the capture-resampling intervals are descriptive and interval overlap is not an equivalence test. The ablation holds the Strong overlay fixed and uses a five-item account/code/CET6 subset. Its $\alpha = 0.2$ value ($N = 135$) is not a repeated estimate of the readability-priority row. Near-full inversion ($\alpha = 1.0$) reduced observed recovery to 18.1% while severely degrading the digital integrated view. Under the 150-frame temporal mean, settings at $\alpha \leq 0.5$ retained 64–73% recovery. The readability-priority choice of $\alpha = 0.2$ is a modeled design decision, not a validated optimum.

D. REAL-CAPTURE VLM PROBES

To test temporal masking against stronger recognizers, we applied three commercially hosted VLM endpoints exposed through the SiliconFlow application programming interface (API): Qwen3-VL-32B-Instruct [47], Kimi-K2.6 [48], and GLM-4.5V [49], thereby showing how recovery varies with model capability on the same photographs.

The probes cover 0° on-axis at three distances. The 60 fps video was split into 12 equal segments and aggregated four ways: (1) *single best*: the single frame within each segment with the highest grayscale contrast; (2) *150-frame mean*: the pixel-wise average of all 150 frames in the segment; (3) *window-mean best*: a sliding 30-frame mean, keeping the window with the highest mean luminance; and (4) *max projection*: the per-pixel temporal maximum across all frames in the segment. The 0.5 m run includes 156 inputs: 36 unprotected short-exposure baselines, 36 high-suppression short-

Unprotected - Digital integrated reconstruction

The quick brown fox jumps over the lazy dog

Unprotected - Short exposure (single frame)

The quick brown fox jumps over the lazy dog

Unprotected - Long exposure

The quick brown fox jumps over the lazy dog

Readability-priority - Digital integrated reconstruction

The quick brown fox jumps over the lazy dog

Readability-priority - Short exposure (single frame)

The quick brown fox jumps over the lazy dog

Readability-priority - Long exposure

The quick brown fox jumps over the lazy dog

High-suppression - Digital integrated reconstruction

The quick brown fox jumps over the lazy dog

High-suppression - Short exposure (single frame)

The quick brown fox jumps over the lazy dog

High-suppression - Long exposure

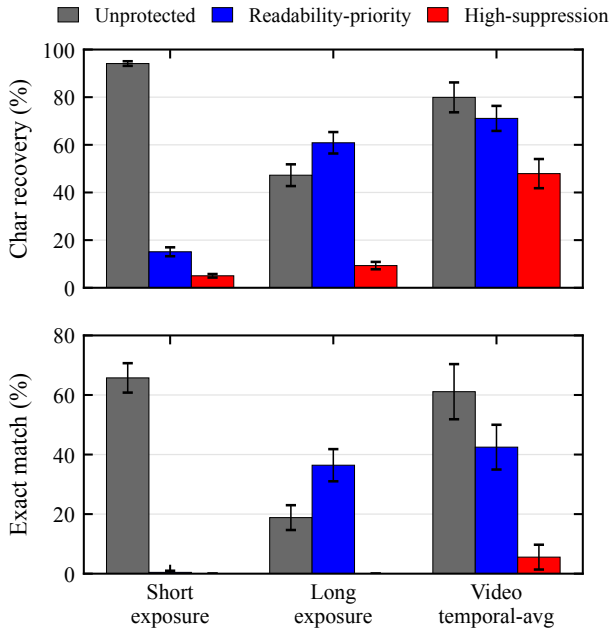
The quick brown fox jumps over the lazy dog

FIGURE 3. Real-capture montage at 1.0 m and 0°. Within each vertically stacked profile group (unprotected, readability-priority, and high-suppression), panels show, from top to bottom, the digital integrated reconstruction, 3.91 ms short-exposure capture, and 31.25 ms long-exposure capture. Camera panels are identically cropped from the archived raw camera JPEGs with no geometric correction or brightness gain, so they differ from the geometrically corrected crops used for OCR evaluation. Digital panels use the same 1920 $\times$ 1080 black playback canvas and brightness-aligned full-cycle integration for each profile (for the unprotected profile this integration equals the displayed source frame); the readability-priority and high-suppression reconstructions include the recorded $\alpha = 0.2$ inversion slot. These digital views are not direct evidence of human readability or visual comfort.

and long-exposure inputs each, and these four aggregate views (12 segments each). The 1.0 m and 1.5 m runs each include 120 high-suppression inputs: 36 short- and long-

TABLE 5. Single-UVC Real-Capture OCR Sensitivity Pool (Best-of-Engine; N = Captures per Row; All 9 Geometries). Brackets Are 95% Capture-Resampling Summaries, Not Population Confidence Intervals. Video Rows Average 150 Frames. The Primary Matched Estimate Is Table 3.

Profile	Attack mode	N	Char recovery [95% summary]	Exact match	Leak rate
Original (unprotected)	Short exposure	324	94.1 [93.1, 95.0]%	65.7%	99.4%
	Video (150-frame mean)	108	79.9 [73.5, 86.0]%	61.1%	86.1%
	Long exposure	324	47.3 [42.8, 51.9]%	18.8%	58.6%
Mask only	Short exposure	324	19.2 [17.0, 21.4]%	0.3%	30.6%
	Video (150-frame mean)	108	79.0 [73.0, 84.9]%	48.1%	88.0%
	Long exposure	324	67.2 [63.0, 71.5]%	35.5%	75.0%
Mask + noise	Short exposure	324	25.6 [22.5, 28.8]%	2.8%	36.1%
	Video (150-frame mean)	108	79.2 [73.3, 85.0]%	49.1%	88.9%
	Long exposure	324	68.0 [63.7, 72.4]%	39.8%	76.2%
Strong (anti-OCR)	Short exposure	324	20.7 [18.2, 23.3]%	0.6%	32.4%
	Video (150-frame mean)	108	78.6 [72.5, 84.5]%	50.9%	88.9%
	Long exposure	324	61.6 [57.1, 66.1]%	39.8%	71.3%
Readability-priority	Short exposure	459	15.1 [13.3, 17.1]%	0.4%	20.5%
	Video (150-frame mean)	153	71.1 [65.7, 76.2]%	42.5%	85.0%
	Long exposure	324	60.9 [56.6, 65.6]%	36.4%	69.1%
High-suppression	Short exposure	324	5.0 [4.3, 5.8]%	0.0%	3.7%
	Video (150-frame mean)	108	47.9 [41.8, 54.0]%	5.6%	76.9%
	Long exposure	324	9.3 [7.9, 10.9]%	0.0%	14.2%

**FIGURE 4.** Best-of-engine OCR character recovery and exact match for three representative real-capture profiles (unprotected, readability-priority, and high-suppression) under short exposure, long exposure, and a 150-frame video temporal mean, pooled across the 9 tested distance-angle configurations. Lower bars indicate less recovered text. The long-exposure and video bars expose the integration boundary: the readability-priority profile retains substantial recoverable content even when its short-exposure bar is low. Readability-priority sample counts include the disclosed five-item repeat and therefore differ from the other profiles.

exposure inputs each plus four video aggregate types (12 segments each). VLM inputs and the OCR best-of-engine (BoE) reference in Table 9 use the same file batch with

TABLE 6. Real-Capture Short-Exposure OCR: Per-Engine Results (Pooled Across 9 Geometric Conditions)

Profile	Engine	Char (%)	Exact (%)	N
Original	Surya	37.1	22.8	324
	EasyOCR	79.5	27.8	324
	Tesseract	84.3	47.8	324
Mask only	Surya	4.2	0.3	324
	EasyOCR	15.6	0.0	324
	Tesseract	3.2	0.0	324
Mask + noise	Surya	11.0	2.8	324
	EasyOCR	18.1	0.0	324
	Tesseract	4.8	0.0	324
Strong	Surya	6.1	0.6	324
	EasyOCR	16.3	0.0	324
	Tesseract	2.1	0.0	324
Readability-priority	Surya	3.2	0.4	459
	EasyOCR	11.7	0.0	459
	Tesseract	3.3	0.0	459
High-suppression	Surya	1.8	0.0	324
	EasyOCR	2.0	0.0	324
	Tesseract	1.9	0.0	324

TABLE 7. Matched Common-Setting Character Recovery Under the Fixed Preprocessing Grid. Each Cell Is Raw Input → Per-Capture Grid Oracle (%); N = 288 per Profile After Duplicate Averaging and Matching.

Engine	Original	Readability	High-supp.
Tesseract	84.9 → 91.8	3.2 → 10.8	2.0 → 8.2
EasyOCR	85.1 → 88.4	14.2 → 33.9	2.3 → 6.5
Surya	30.6 → 49.3	4.1 → 20.2	2.0 → 6.0
Best of three	94.5 → 95.9	17.8 → 40.2	5.6 → 13.7

nominal 3.91/31.25 ms labels. All VLM calls use raw camera JPEGs.

Of 1,188 model requests, 1,133 returned parseable results:

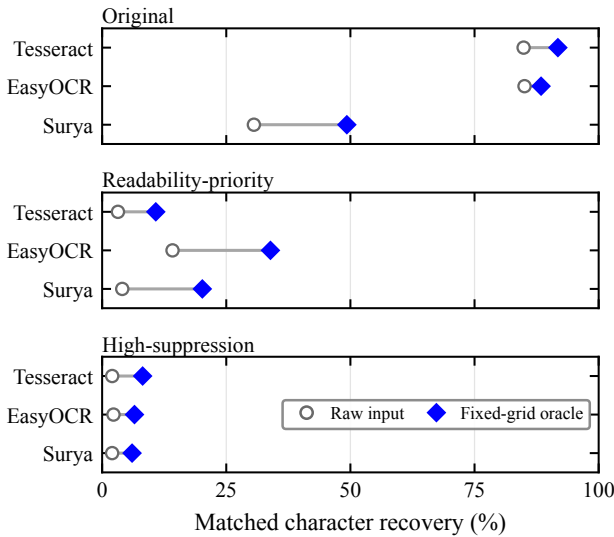


FIGURE 5. Raw-input versus fixed-grid-oracle character recovery for each OCR engine on the matched common-setting pool ($N = 288$ per profile). The oracle selects the best of raw plus five predeclared transforms separately for each capture and metric. Lines show within-engine increases, not uncertainty intervals. Exact values appear in Table 7.

TABLE 8. Inversion Frame Strength α Ablation on the Fixed Strong Overlay ($\alpha_s = 0.10$, $\alpha_g = 0.12$): Real-Capture Best-of-Engine Character Recovery (Mean and 95% Capture-Resampling Interval, Five Account/Code/CET6 Items Pooled Across 9 Geometric Conditions; $N = 135$ per Setting for Long Exposure, $N = 45$ for Video Temporal Averaging). This Five-Item Subset Differs from the 12-Item Readability-Priority Rows in Table 5.

α	Long-exp. char (%)	Video avg. char (%)
0.0	68.6 [63.1, 73.9]	72.7 [63.3, 81.1]
0.2	67.0 [61.3, 72.7]	69.8 [59.9, 78.7]
0.3	72.0 [66.2, 76.9]	64.3 [54.0, 73.6]
0.5	61.7 [55.6, 67.7]	66.4 [56.5, 75.2]
1.0	18.1 [14.0, 22.0]	28.4 [20.5, 36.9]

34 failures at 0.5 m, 21 at 1.0 m, and none at 1.5 m. Each table cell reports effective N and is conditional on a parseable response. GLM failures concentrate on mixed, digit, and credential items, so missingness cannot be treated as random. We computed cell-wise lower and upper bounds by assigning every failed call 0 or 100% recovery. For the GLM 0.5 m short cell, character recovery is bounded by 61.0–74.9% and exact recovery by 44.4–58.3%. Conditional cells are not used to rank models; complete bounds for all cells are archived with the analysis.

We use the failure-free 1.5 m run as primary cross-model evidence; the 1.0 m run enters the short-exposure distance sweep as supplementary data. At 1.0 m, Qwen and Kimi long-exposure exact match are 84.4% and 74.3%, and the successful calls from the three models yield 70.0–83.3% exact match under the 150-frame video mean. GLM long-exposure exact match is only 3.3% at 1.0 m versus 58.3% at 1.5 m, but content-dependent failures preclude a cross-distance consistency claim. For compactness, Table 9 reports only the 150-

frame video mean at 0.5 m; the 1.5 m run reports all four aggregate views to show view-selection sensitivity.

All calls used SiliconFlow’s OpenAI-compatible API. The recorded provider identifiers are Qwen/Qwen3-VL-32B-Instruct, Pro/moonshotai/Kimi-K2.6, and zai-org/GLM-4.5V. Qwen3-VL and GLM-4.5V have public technical reports [47], [49]. For Kimi-K2.6, the public model card [48], the provider endpoint identifier, and archived request metadata form the reproducibility record. Decoding temperature is 0 and maximum output is 4,096 tokens. The shared prompt requests only visible text and prohibits guessing; ground truth is not provided. Full prompts, parameters, and per-call outputs are archived. Metrics follow §V-A and are pooled at the capture level.

0.5 m session selection transparency: The archive preserves an alternative session (`real_capture_vlm_d0.5_a0_rerun.json`, timestamp 012715) at the same position whose high-suppression short-exposure inputs are nearly all-black, yielding empty transcriptions and 0% char/exact across all three models. Table 9 uses the 141107 (3.91 ms) session because it shares the OCR reference batch and its inputs are not black. These are the only two 0.5 m on-axis VLM sessions, so Qwen’s observed short-exposure exact-match outcome ranges from 0/36 captures in the underexposed session to 28/36 captures in the tabled session, not a stable single-session estimate. The adequate-exposure criterion was applied after inspecting two sessions and is therefore a post-hoc selection rule; preserving both sessions and reporting the range makes the exposure sensitivity explicit. The tabled session is the conservative attacker-favorable case, while the alternative is retained as an underexposure diagnostic.

Table 9 presents the pooled VLM probe results, where short-exposure rows show distance variation while long-exposure and video aggregate rows expose integration attack boundaries; Table 10 breaks down 0.5 m high-suppression short-exposure character recovery by content type.

Key findings:

- VLMs recover substantially more text than conventional OCR:** On the same 0.5 m photographs, best-of-engine OCR yields 8.4% char / 0% exact on high-suppression short exposure, while Qwen3-VL achieves 91.5% char and exactly recovers 28/36 captures. At 1.5 m, the three models retain 33.3–47.2% exact / 61.5–70.0% char. The gap between conventional OCR and modern VLMs is large.
- VLM recovery concentrates on large-font short snippets:** At 0.5 m, CET6 document pages yield 0.4–18.5% VLM character recovery, whereas credentials, code, and digit strings yield 50–100% (Table 10). At 1.5 m, CET6 pages drop to 0% across the three models, while digit strings and short sentences remain recoverable. The pattern is consistent with font scale and language priors [50], [51].
- Attack capability varies across models and conditions:** In the failure-free 1.5 m session, nominal long-exposure exact recovery ranges from 58.3% to 91.7%,

TABLE 9. Real-Capture VLM Probes (0° On-Axis; All Rows Except the Original Short Baseline Are High-Suppression). Each VLM Cell Reports Exact Count / Char / N (Recoveries / % / Effective Calls); Cells with Failures Are Conditional on Parseable Responses. OCR BoE Shows the Same-Batch Three-Engine Best-of-Engine Reference (Exact / Char, %). The Temporal-Mean Rows Average 150 Short-Exposure Frames; Other Aggregate Views Are Defined in the Text. The 0.5 m High-Suppression Short Row Uses the Adequate-Exposure Attacker-Favorable 141107 Session; the Only Alternative 0.5 m Session Was Underexposed and Yielded 0/36 Qwen Exact Matches.

Distance	Condition	OCR BoE	Qwen3-VL-32B	Kimi-K2.6	GLM-4.5V
<i>Short-exposure distance sweep</i>					
0.5 m	original short	58.3 / 92.4	30 / 94.5 / 36	30 / 95.6 / 36	19 / 92.0 / 26
0.5 m	short	0.0 / 8.4	28 / 91.5 / 36	17 / 84.7 / 36	16 / 70.8 / 31
1.0 m	short	0.0 / 3.6	20 / 79.1 / 36	14 / 68.6 / 36	9 / 47.0 / 33
1.5 m	short	0.0 / 6.5	17 / 68.9 / 36	12 / 61.5 / 36	13 / 70.0 / 36
<i>Key attack views</i>					
0.5 m	long	0.0 / 2.1	28 / 85.7 / 34	1 / 7.9 / 36	0 / 4.9 / 30
0.5 m	Video (150-frame mean)	8.3 / 50.7	10 / 94.0 / 12	10 / 95.5 / 12	7 / 91.8 / 8
1.5 m	long	0.0 / 13.2	33 / 89.3 / 36	32 / 89.9 / 36	21 / 62.4 / 36
1.5 m	Video (single best)	0.0 / 4.3	1 / 24.8 / 12	0 / 12.7 / 12	0 / 29.8 / 12
1.5 m	Video (150-frame mean)	0.0 / 48.4	8 / 90.3 / 12	8 / 90.3 / 12	7 / 88.5 / 12
1.5 m	Video (window-mean best)	0.0 / 8.6	7 / 89.0 / 12	5 / 84.2 / 12	8 / 87.2 / 12
1.5 m	Video (max projection)	0.0 / 19.0	8 / 90.1 / 12	7 / 89.2 / 12	7 / 88.7 / 12

TABLE 10. 0.5 m High-Suppression Short-Exposure VLM Character Recovery by Content Type. Baseline Column Shows Same-Session Unprotected Original Short Results for Qwen (Kimi/GLM Similar); GLM Had Occasional Call Failures, Parenthesized N Shows Effective Count. At 1.5 m, CET6 Dense Pages Yield 0% Char Across All Three Models.

Content type (N)	Baseline (%)	Qwen (%)	Kimi (%)	GLM (%)
CET6 document (3)	64.4	18.5	5.6	0.4
Credentials (6)	97.6	97.6	88.3	77.1 (5)
Code snippets (6)	89.8	94.9	95.8	88.9
Digit strings (6)	100.0	100.0	98.6	50.0 (4)
Mixed content (6)	100.0	100.0	92.5	98.2 (4)
Paragraphs (6)	100.0	99.8	81.7	66.5
English sentences (3)	95.3	95.3	96.9	94.6

with 62.4–89.9% character recovery. Endpoint, distance, capture view, and content jointly shape this spread.

- 4) **Sustained video aggregation strongly increases VLM recovery:** At 0.5 m, missingness bounds for the 150-frame mean span 58.3–91.7% exact and 61.2–95.5% character recovery across models. In the failure-free 1.5 m session, the 150-frame mean yields 58.3–66.7% exact and 88.5–90.3% character recovery, while single-best-frame exact recovery is 0–8.3%. The accompanying 12.7–29.8% single-best character values rest on low-information inputs, where fluent language-prior completions can inflate character recovery even though the prompt prohibits guessing; the exact-match counts are the more conservative reading.

E. USER EXPERIENCE STUDY

To evaluate the impact of the protected display on human readability and visual comfort, we designed and conducted a web-based user study.

Equipment and environment protocol: The same 240 Hz display as in §V-A. The formal study version sets a hard measured refresh-rate threshold of ≥ 200 Hz to ensure the

fixed $n = 4$ basic subframe cycle rate $f_r/n \geq 50$ Hz; below this threshold participation is blocked rather than silently reducing n . The 144–199 Hz adaptive path is available only for an explicitly labeled demonstration mode, and its sessions are excluded from analysis by default. Before each participant begins, the experimenter confirms fixed display brightness, disabled auto-brightness/power-saving and variable refresh rate, unified browser fullscreen, closed high-load background processes, and an approximate 60 cm viewing distance. Refresh rate is re-measured after any display-mode change.

The browser records actual frame intervals in each trial's `requestAnimationFrame` loop; if the interval between adjacent callbacks exceeds $1.5\times$ the expected interval computed from the pre-trial measured refresh rate, a dropped-frame event is logged, along with estimated drop count, drop rate, mean/max frame intervals, observed refresh rate, basic subframe cycle rate, and full cycle rate with inversion frame. This recording reflects only browser presentation cadence and cannot substitute for panel scanning or photometric measurement.

Ethics and safety: The study is minimal-risk human-subjects research. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type, so no institutional protocol number exists. Every session was administered in person by the experimenter under the safeguards described here. Before each session, the experimenter individually confirmed the participant's voluntary consent and photosensitivity-screening response in person. The web page then reiterated that the masked display involves rapid temporal modulation, that participants should stop immediately upon experiencing discomfort, eye fatigue, dizziness, nausea, or headache, and that they could withdraw unconditionally at any time. Proceeding recorded the confirmation timestamp and screening flag. Individuals who reported a history of photosensitive epilepsy or sensitivity to flicker were not eligible to continue. The implementation

applies a conservative $f_r/n \geq 50$ Hz operating floor, but this engineering gate is not an ethical or clinical safety determination. At $n = 4$ on a 240 Hz panel the basic cycle rate is 60 Hz; the inversion frame reduces the full cycle frequency to $f_r/(n + 1) = 48$ Hz, and the implementation separately logs warnings and actual timing. The formal study did not collect names, student identifiers, majors, ages, or gender information. Analysis and publication use only de-identified study records. Vision-correction status was collected to characterize the analyzed sample.

Task design: Each participant first completes one 12-second, unscored unmasked Canvas typing warm-up, views a 10-second, unscored readability-priority full-mask preview, and then completes an 8-second, unscored typing practice under the same readability-priority mask. This sequence separates initial stimulus and input adaptation from the four subsequent 20-second scored trials (within-subjects design). On every typing screen, including warm-up, practice, and scored trials, the stimulus Canvas is visible and its player has started before the participant clicks “Start.” Clicking “Start” triggers a 5-second countdown while the input remains disabled; the timed typing interval and input availability begin only when the countdown ends. The same pre-start viewing procedure applies to both control and masked conditions.

The two conditions are each repeated twice. The experimenter’s continuously assigned zero-based sequence number k selects ABBA or BAAB order to counterbalance practice and fatigue: (A) **original text** (control), displayed as unmasked Canvas with $n = 1$; (B) **masked text**, using the readability-priority configuration: $n = 4$, mask+noise, strong anti-OCR ($\alpha_s = 0.10$, $\alpha_g = 0.12$, 6 cycles) + weak inversion ($\alpha = 0.2$). The typing order index is $k \bmod 2$, and the subjective-rating Latin-square row is $\lfloor k/2 \rfloor \bmod 6$. Every consecutive 12 participants therefore cover all 2×6 joint assignments. Formal sessions enforce a uniqueness constraint on k , and the backend recomputes and verifies both order indices.

Both conditions are rendered by the same `render SourceCanvas` path, with identical dark background, light text, canvas size, 23 px bold font, and color. The sole processing difference is the temporal protection pipeline, avoiding polarity and font-weight confounds. Here “cycles” means that the web implementation rotates through six distinct mask, noise, and stripe instances before repeating. This web-playback-specific setting shares α_s and α_g with the real-capture configuration in §IV and avoids repeatedly integrating one fixed overlay pattern. The web player is the independent JavaScript/Canvas2D path described in §IV: deterministic seeded masking, pixel-space gain, and surrogate texture noise, rather than the ChaCha20- and gradient-based Python pipeline. Frame-scheduling logs verify presentation timing at the JavaScript level; no photometric measurement of per-subframe panel output was performed.

Scoring no longer compares characters position-by-position but aligns the full transcription against the optimal target prefix using Levenshtein minimum string distance

(MSD) [52]; untyped target suffix does not count as error. The alignment yields accuracy, characters per minute (CPM), and words per minute (WPM), along with edit distance, aligned prefix length, and scoring version. First-keystroke latency is defined from the end of the 5-second countdown, when the input is enabled and the scored timer starts, to the first input event that makes the transcription non-empty. Analysis first averages each participant’s two trials per condition, then performs within-subject paired comparison.

Subjective ratings: After the typing trials, participants rate 6 conditions on the three 1–5 Likert items displayed by the interface: readability (1 = difficult to read, 5 = very clear), stability (stored in the `flicker` field; 1 = strong flicker, 5 = no perceptible flicker), and immediate visual comfort (fatigue; 1 = very uncomfortable, 5 = very comfortable). Higher values therefore indicate a better outcome on every item. Conditions include the unmasked original anchor, $n \in \{2, 3, 4\}$ mask+noise, $n = 4$ mask-only, and the same $n = 4$ readability-priority full configuration (mask+noise+strong anti-OCR+weak inversion) used in the typing task. The first three masking levels all achieve ≥ 60 Hz real temporal playback on a 240 Hz panel; the originally planned $n = 8$ would yield 30 Hz and trigger static fallback, so it is not included as a temporal ablation. The six conditions are presented in a 6th-order balanced Latin square; each condition must be viewed for at least 10 seconds before submission is enabled, with viewing start/end times and duration recorded. The unmasked anchor is also an attention/scale-interpretation aid but is not mechanically excluded by a “must rate 5” rule. This study does not include the high-suppression profile; assessment of its visible stripes and grain’s acceptability is left to future work (see §VI).

Sample size and analysis plan: A minimum of $N = 24$ participants was planned, a number within the range required for paired effect sizes of $d_z \approx 0.6$ – 0.8 at 80% power, and exactly covering two rounds of the 12-person joint order assignment. Pre-specified exclusion criteria are debug/demo sessions, incomplete four typing or six rating records, measured refresh rate below 200 Hz, any typing trial with fewer than 5 attempted characters, control mean accuracy below 50%, masked trials not in temporal mode, any trial with observed basic cycle rate below 50 Hz, or minimum-view straight-line ratings. The last criterion is defined as all six rating rows having a recorded viewing duration no greater than 11 seconds (a 1-second tolerance above the 10-second submission gate) and, within every row, identical scores on all three items. WPM, CPM, accuracy, attempted characters, and first-keystroke latency are first averaged within each participant by condition; speed metrics must be interpreted jointly with accuracy to avoid misinterpreting fast input under low accuracy as performance improvement. Paired differences are tested using paired t -tests when the pre-specified Shapiro test is passed, otherwise Wilcoxon signed-rank tests, with effect sizes and participant-level bootstrap 95% confidence intervals reported. Likert data are analyzed per dimension using Friedman tests with Holm-corrected post-hoc pairwise

TABLE 11. Typing Task: Within-Subject Comparison of the Control and Readability-Priority Masked Conditions ($N = 56$; Participant Means Over Two Trials per Condition; * = Masked – Control With Participant-Level Bootstrap 95% CI). Per Metric, a Paired t -Test Is Used When the Pre-Specified Shapiro Test Passes (Effect Size Cohen's d_z), Otherwise a Wilcoxon Signed-Rank Test (Effect Size Rank-Biserial r_{rb}); Test Statistics Are Reported in the Text

Metric	Control	Masked	Δ [95% CI]
Accuracy (%)	96.54	94.83	-1.71 [-3.39, -0.04]
WPM	20.44	19.30	-1.14 [-1.91, -0.39]
CPM	102.19	96.48	-5.71 [-9.56, -1.88]
Attempted chars	35.21	33.47	-1.74 [-2.88, -0.62]
First-key latency (ms)	742.64	833.61	+90.96 [-31.11, +210.69]

TABLE 12. Subjective Ratings for the Six Display Conditions ($N = 56$; 1–5 Likert, Higher Is Better; Mean $\pm$ SD). Stability Is the Inverse-Flicker Item and Comfort the Immediate Visual Comfort Item (§V-E); Friedman and Holm-Corrected Pairwise Wilcoxon Statistics Are Reported in the Text

Condition	Readability	Stability	Comfort
Unmasked anchor	4.80 $\pm$ 0.64	4.79 $\pm$ 0.68	4.75 $\pm$ 0.61
$n=2$ mask+noise	4.61 $\pm$ 0.68	4.77 $\pm$ 0.57	4.48 $\pm$ 0.87
$n=3$ mask+noise	4.21 $\pm$ 0.87	4.57 $\pm$ 0.76	4.11 $\pm$ 0.91
$n=4$ mask+noise	4.23 $\pm$ 0.91	4.38 $\pm$ 0.93	4.02 $\pm$ 1.05
$n=4$ mask-only	4.16 $\pm$ 0.87	4.45 $\pm$ 0.78	4.02 $\pm$ 0.96
Readability-priority full ($n=4$)	3.79 $\pm$ 1.17	2.98 $\pm$ 1.30	3.39 $\pm$ 1.25

Wilcoxon tests. Analysis scripts generate de-identified participant means, JSON audit reports, and two $\text{\LaTeX}$ tables directly from the formal database `study_formal.db`.

Participants: 61 volunteers were recruited; 56 sessions met all pre-specified inclusion criteria and entered analysis (5 excluded: 3 for minimum-view straight-line ratings and 2 for control mean accuracy below 50%; the per-criterion audit accompanies the released analysis output). The 61 formal sessions filled each of the twelve joint order buckets five times, with one bucket receiving a sixth session. The formal study did not collect names, student identifiers, majors, ages, or gender information, so no age or gender distribution is reported; 46 of the 56 wore glasses, 1 wore contact lenses, and 9 reported no vision correction. All formal sessions ran on the 240 Hz laboratory display with a measured refresh rate of at least 200 Hz.

Results: Table 11 and Fig. 6 summarize the typing task. Relative to the control condition, the readability-priority full configuration changed mean accuracy by -1.71 percentage points (95% CI [-3.39, -0.04], Wilcoxon $W = 428$, $p = 0.043$, $r_{rb} = -0.33$), WPM by -1.14 (95% CI [-1.91, -0.39], $t(55) = -2.93$, $p = 0.005$, $d_z = -0.39$), and first-keystroke latency by $+91.0$ ms (95% CI [-31.1, +210.7], Wilcoxon $W = 641$, $p = 0.200$, $r_{rb} = 0.20$); CPM and attempted characters gave $p = 0.005$ ($d_z = -0.39$) and $p = 0.004$ ($d_z = -0.40$), respectively. Under Holm correction across the five typing metrics, WPM, CPM, and attempted characters remain significant whereas the accuracy difference ($p = 0.043$, rank 4 of 5) does not. Table 12 and Fig. 7 summarize the subjective ratings: Friedman tests yielded $\chi^2(5) = 72.2$ ($p = 3.5 \times 10^{-14}$)

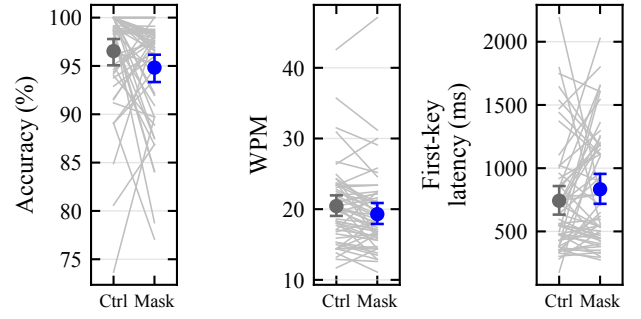


FIGURE 6. Within-subject typing performance under the control and readability-priority masked conditions: each gray line is one participant (mean of two trials per condition); markers show group means with participant-level bootstrap 95% confidence intervals. Panels show transcription accuracy, words per minute, and first-keystroke latency.

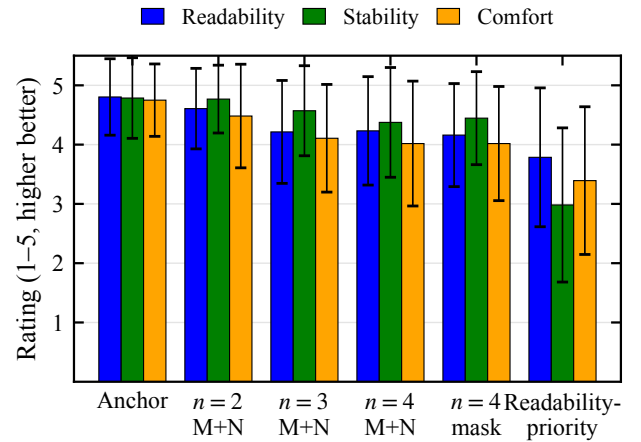


FIGURE 7. Mean 1–5 Likert ratings (error bars: SD) for readability, stability, and immediate visual comfort across the six rated display conditions, ordered from the unmasked anchor to the readability-priority full configuration. Higher is better on every item.

for readability, $\chi^2(5) = 123.2$ ($p = 6.5 \times 10^{-25}$) for stability, and $\chi^2(5) = 98.5$ ($p = 1.1 \times 10^{-19}$) for immediate visual comfort; Holm-corrected pairwise comparisons between the readability-priority full configuration and the unmasked anchor gave $p = 2.6 \times 10^{-4}$, $p = 2.2 \times 10^{-7}$, and $p = 1.8 \times 10^{-6}$ on the three items, respectively.

F. REAL-CAPTURE OBJECT DETECTION AND TRACKING

To probe whether the temporal degradation transfers beyond text, we report an exploratory detection/tracking stress test within the same camera link, noting that the evidence strength is not equivalent to the OCR main experiment. Content was displayed on the 240 Hz screen, captured by the S600 at $1.5 \text{ m} / 0^\circ$, and processed by four detectors [53]–[56]. mAP, mAP50, and average recall (AR) are reported for detection. Real-capture tracking uses higher-order tracking accuracy (HOTA) and identification F1 (IDF1); simulated tracking additionally reports multiple-object tracking accuracy (MOTA) and multiple-object tracking precision (MOTP). The pro-

TABLE 13. Real-Capture COCO Detection (150 Images; *real_clean* = Unprotected Capture Baseline)

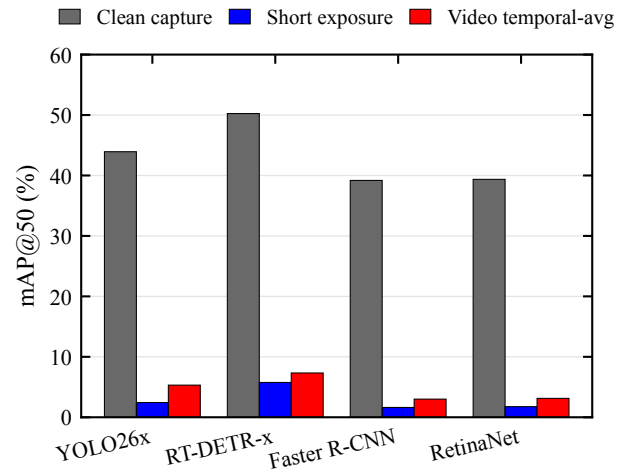
Detector	Condition	mAP	mAP50	AR
YOLO26x	<i>real_clean</i>	28.6%	43.9%	31.0%
	<i>real_short</i>	1.3%	2.4%	1.4%
	<i>real_video</i>	3.5%	5.3%	3.6%
RT-DETR-x	<i>real_clean</i>	30.1%	50.2%	34.8%
	<i>real_short</i>	3.5%	5.8%	4.1%
	<i>real_video</i>	4.6%	7.3%	5.5%
Faster R-CNN	<i>real_clean</i>	21.3%	39.2%	28.7%
	<i>real_short</i>	0.8%	1.6%	0.9%
	<i>real_video</i>	1.5%	3.0%	1.8%
RetinaNet	<i>real_clean</i>	22.6%	39.4%	27.0%
	<i>real_short</i>	0.9%	1.8%	1.0%
	<i>real_video</i>	1.9%	3.1%	2.0%

TABLE 14. Real-Capture MOT17-09-FRCNN Still-Frame Tracking (ByteTrack-Style Greedy Association, 450 Frames; HOTA and IDF1 Computed by TrackEval [58])

Detector	Condition	HOTA	IDF1
YOLO26x	<i>real_clean</i>	15.8%	12.4%
	<i>real_short</i>	8.7%	6.5%
	<i>real_video</i>	10.0%	8.7%
RT-DETR-x	<i>real_clean</i>	14.2%	10.5%
	<i>real_short</i>	8.1%	5.4%
	<i>real_video</i>	9.6%	6.9%
Faster R-CNN	<i>real_clean</i>	14.4%	9.8%
	<i>real_short</i>	6.8%	6.2%
	<i>real_video</i>	10.5%	7.6%
RetinaNet	<i>real_clean</i>	14.2%	10.3%
	<i>real_short</i>	5.1%	4.4%
	<i>real_video</i>	6.8%	5.9%

tected condition uses the $n = 4$ mask+noise profile without overlays or inversion. The *real_clean* reference is an unprotected nominal 31.25-ms capture. The *real_short* image is the maximum-grayscale-contrast frame from a three-frame protected burst at the nominal 3.91-ms setting, and *real_video* is the mean of 150 protected frames acquired nominally at 60 fps. Tracking uses 450 frames from MOT17-09-FRCNN via still-display-then-capture and an in-project greedy association implementation inspired by ByteTrack [57].

Under short exposure, all four detectors' mAP50 drops from 39.2–50.2% to 1.6–5.8% (Table 13, Fig. 8), representing relative decreases exceeding 85%, while the *real_video* condition yields 3.0–7.3%, also below the *real_clean* baselines. Because protection, exposure, resolution, and frame reduction differ across conditions, these values form a cross-task stress test rather than an isolated protection effect. In still-frame tracking (Table 14), HOTA decreases from 14.2–15.8% to 5.1–8.7% and IDF1 from 9.8–12.4% to 4.4–6.5%. Detection mAP was computed by pycocotools; tracking HOTA and IDF1 were computed by TrackEval [58]. The tracker is an in-project greedy IoU association implementation.

**FIGURE 8.** Real-capture mAP50 for four detectors across the unprotected capture, short-exposure protected capture, and 150-frame protected mean. Conditions differ in protection, exposure, resolution, and frame reduction.**TABLE 15.** Synthetic Single-Subframe OCR: Three-Engine Character Recovery (120 Samples, $n = 4$, $\varepsilon = 8/255$)

OCR engine	Original frame	Single subframe	Reduction
Tesseract	94.0%	0.0%	94.0%
EasyOCR	94.1%	0.0%	94.1%
Surya	97.0%	2.6%	94.4%

G. SOFTWARE SIMULATION EXPERIMENTS

This section uses software simulation to address two questions: (1) what is the machine readability of one protected subframe obtained after the application renderer, and (2) what is recovery under zero camera noise and perfect alignment? By characterizing the post-render single-subframe capture point, the experiment defines a boundary that malware intercepting the original frame or accumulating a complete cycle would bypass.

1) Synthetic Single-Subframe OCR

Using Tesseract [46], EasyOCR [45], and Surya [44] on 120 synthetic samples, character recovery drops from 94.0–97.0% to 0–2.6% across three engines (Table 15). Stratified breakdowns further characterize recovery across the evaluated content categories.

Fig. 9 visualizes these results; single-subframe recovery does not exceed 2.6% for any engine.

2) Theoretical Security Boundary

When an attacker accumulates registered frames covering a full cycle, complementary subframes can be re-summed in the ideal linear model of (1), with the archived per-sample strongest digital attack recovering 95.2% and pure full-cycle temporal averaging recovering 94.3% (Table 16)—values that empirically mark the reconstruction boundary of the evaluated linear decomposition. The high-suppression profile

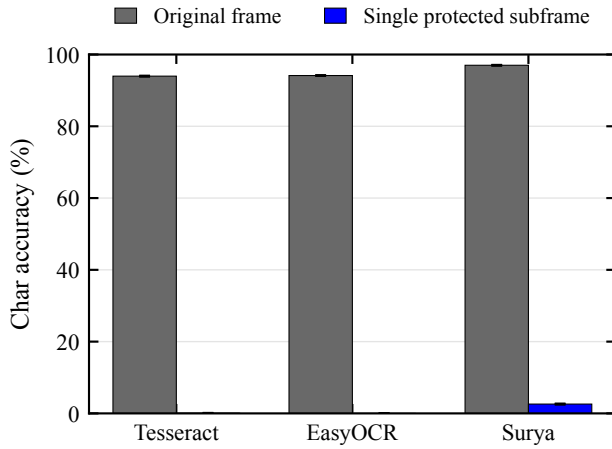


FIGURE 9. Character recovery for three OCR engines on original frames and single protected subframes; error bars are 95% bootstrap intervals resampled by corpus sample.

TABLE 16. Archived Digital Reconstruction Boundary (Tesseract)

Setting	<i>N</i>	Char.	Exact	SSIM	ΔE_{00}
Base: cycle mean	120	94.3%	85.8%	–	–
Base: per-sample strongest	120	95.2%	91.7%	–	–
Readability: cycle mean	16	87.9%	50.0%	0.948	1.46
High: cycle mean	16	56.6%	6.3%	0.764	4.87
High: inversion slot	16	93.0%	81.2%	–	–

The 120-sample base attack and 16-sample profile ablation are separate archived digital evaluations. “Per-sample strongest” selects the best result among seven prespecified reconstructions (temporal average cycle, blue channel max, differential blue, differential luma, global shutter slot 0, phase search max, and phase search mean) for each sample. “Inversion slot” extracts only the $\alpha(1 - I)$ frame and applies contrast stretching; its 93.0% recovery identifies slot-level composition as a critical design variable. Profile SSIM and ΔE_{00} describe full-cycle reconstruction quality, not physical or user-perceived quality.

yields 47.9% pooled recovery under the 150-frame mean and 53.9% after excluding the underexposed position. Character recovery also depends on content and the physical imaging path.

3) Detection and Tracking Simulation

Simulations were run on COCO val2017 [59] and MOT17 [60] with YOLO26x [53], RT-DETR-x [54], Faster R-CNN [55], and RetinaNet [56]: the COCO simulation used 8 images as an implementation pipeline diagnostic (Table 17), while the MOT simulation covers 7 sequences totaling 5,316 frames and uses the same in-project ByteTrack-inspired greedy association method (Table 18).

On the 8 COCO images, single-subframe mAP50 across the four detectors is 0–6.0%. Across the 5,316 MOT frames, single-subframe IDF1 is 0.1–5.4%, while MOTA includes negative and nonmonotonic values. Negative MOTA occurs when the false-positive, false-negative, and identity-error penalties exceed the number of ground-truth objects; it is not a negative detection probability. Detector architecture, confidence thresholds, proposal density, and the greedy associator jointly shape the observed differences. Fig. 10 summarizes

TABLE 17. COCO val2017 Simulated Detection Pipeline Diagnostics (4 Detectors, 8 Images)

Detector	Condition	mAP	mAP50	AR
YOLO26x	clean	48.1%	58.3%	51.0%
	single subframe	0.0%	0.0%	0.0%
	temporal avg.	14.3%	18.3%	14.3%
RT-DETR-x	clean	47.1%	58.7%	53.4%
	single subframe	5.2%	6.0%	5.2%
	temporal avg.	16.8%	23.7%	17.9%
Faster R-CNN	clean	38.2%	52.7%	43.6%
	single subframe	3.0%	4.4%	3.0%
	temporal avg.	11.1%	16.5%	11.1%
RetinaNet	clean	33.2%	47.1%	35.3%
	single subframe	3.3%	4.2%	3.3%
	temporal avg.	9.6%	13.2%	9.6%

TABLE 18. MOT17 Simulated Tracking (ByteTrack-Style Greedy Association; 5,316 Frames; CLEAR and IDF1 by In-Project Approximate Implementation)

Detector	Condition	MOTA	MOTP	IDF1
YOLO26x	clean	31.1%	81.7%	27.1%
	single subframe	0.1%	74.8%	0.1%
	temporal avg.	9.1%	78.1%	12.0%
RT-DETR-x	clean	24.1%	79.8%	38.2%
	single subframe	–2.3%	73.2%	5.4%
	temporal avg.	–11.4%	75.0%	14.7%
Faster R-CNN	clean	3.8%	77.9%	35.6%
	single subframe	–2.5%	71.2%	4.5%
	temporal avg.	–2.7%	73.2%	14.0%
RetinaNet	clean	–5.4%	77.1%	30.5%
	single subframe	0.4%	70.6%	2.7%
	temporal avg.	–2.9%	73.8%	11.9%

the residual capability of a single protected subframe relative to the clean baseline across the evaluated OCR, detection, and tracking pipelines.

VI. DISCUSSION

A. PRINCIPAL FINDINGS AND RESEARCH VALUE

At the common S600 setting, readability-priority recovery is 17.8%, compared with 94.5% for unprotected content. High-suppression further lowers recovery to 5.6%. The paired content-cluster gaps remain 76.7 and 88.9 percentage points across 288 matched units per profile. A reproducible preprocessing grid raises protected recovery, yet large paired gaps remain. Detection, tracking, VLM, integration, and simulation experiments then map how attacker capability changes these outcomes.

Relative to prior temporal-display studies, the 10,575-capture physical archive is new, as is the link from average OCR recovery to explicit-field leakage and upper-tail risk and the evaluation of attacker escalation across preprocessing, sustained capture, and modern recognition models.

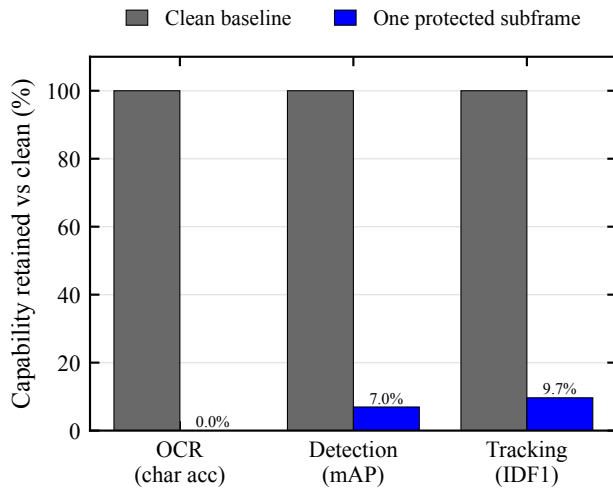


FIGURE 10. Residual capability of one protected subframe relative to the clean baseline (clean = 100%) across the evaluated OCR, detection, and tracking pipelines. The metrics are normalized within task and are not directly commensurate across tasks.

B. INTEGRATION AND ATTACKER ESCALATION

Temporal masking exploits the difference between human temporal integration and short camera exposure; in the ideal linear model, registered complementary subframes can reconstruct the original because $\sum \mathbf{I}_k = \mathbf{I}$, and the digital attack evaluation confirms this boundary, reaching 95.2% per-sample strongest recovery and 94.3% under pure full-cycle averaging.

Physical temporal averaging also increases recovery without requiring the mask seed or phase: the 150-frame mean reaches 71.1% for readability-priority and 47.9% for high-suppression, rising to 79.7% and 53.9% at the common settings, so that sustained capture converts temporal fragments into a usable recognition signal.

C. PROFILE TRADE-OFFS AND PRACTICAL IMPLICATIONS

The two principal profiles occupy distinct operating points: readability-priority preserves a digitally reconstructed SSIM of 0.948 but retains 17.7% explicit-field recovery, whereas high-suppression lowers several recovery measures while reducing digital SSIM to 0.764 and increasing ΔE_{00} from 1.46 to 4.87, making the privacy-readability trade-off between these profiles explicit and tunable.

This trade-off runs opposite to Kaleido-class temporal re-encoding [10]: Kaleido relies on perceptually complete mixing within a cycle to preserve viewing experience, so full-cycle integration reconstructs an approximation of the original, whereas the composite profiles here give up completeness because, for static content, completeness is the entry point for integration attacks.

Because the high-suppression inversion slot reaches 93.0% recovery after contrast stretching, slot-level composition remains a weak point in the current design, and a practical system should therefore combine temporal masking with field-

level policies and independent access controls for high-value content.

D. VLM-AWARE IMPLICATIONS

Modern VLMs recover far more text than conventional OCR from the same protected photographs, with recovery strongest for large-font credentials, code, digits, and short sentences while dense CET6 pages remain more strongly suppressed. The implication is that temporal profiles should adapt to glyph scale, semantic predictability, and field sensitivity. Although high-suppression strongly reduces dense small-text recovery under conventional OCR, VLMs still recover many large-font credentials, code snippets, and digit strings; content-adaptive, VLM-aware temporal profiles warrant investigation.

E. ETHICS STATEMENT

This research is defensive in nature, aiming to protect screen content from unauthorized photography. The attack analysis (§V-G2) is intended for honest assessment of security boundaries, not as an attack guide. The user study involved human subjects in minimal-risk behavioral research. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type, so no institutional determination was available. Before each session, the experimenter individually confirmed voluntary consent and photosensitivity-screening eligibility in person; the web interface then recorded the confirmation state and timestamp. The protocol also included in-person supervision and voluntary withdrawal as described in §V-E. Analysis and publication use only de-identified data.

F. LIMITATIONS AND FUTURE WORK

The physical evidence comes from one S600 UVC webcam, and profile order was fixed rather than randomized. Exposure values are control-derived labels rather than photometric shutter measurements. The archive also lacks decoded auto-exposure state, gain, white-balance, focus, ambient illuminance, display-camera phase, and physical panel luminance. These factors can interact with time, geometry, and profile order.

The current PoC operates at the application layer with fixed panel output. It lacks photometric slot-timing measurements, luminance-matched static controls, and matched noise-off versions of the enhanced profiles. Temporal sparsity, duty-cycle luminance, overlays, panel response, camera exposure, and image signal processor (ISP) processing therefore remain coupled. Rolling-shutter row mixing and panel transitions are candidate explanations for the simulation-to-real gap.

The content-cluster intervals hold the tested geometries fixed, and the 12-item corpus is too small for script-level inference. Detection uses 150 COCO images at one position, while tracking uses 450 still-frame captures from one MOT17 sequence. The VLM probes use one on-axis profile, three API endpoints, and raw JPEG inputs. Commercial OCR, learned restoration, and unconstrained adaptive preprocessing may

recover additional text. The 150-frame experiment also leaves the duration–recovery curve unresolved.

Future work should randomize profile order, lock camera parameters, record batch time, and measure physical slot timing and luminance. Matched static controls can separate temporal and luminance effects. Replication across cameras and displays can test transfer, while continuous-video attacks can map recovery against recording duration. VLM-aware profiles can explore font-size-adaptive granularity, glyph decoys, and field-level rendering. Extending the usability study to longer sessions, other panels, and the high-suppression profile would map comfort beyond the present short-duration evaluation.

VII. CONCLUSION

We evaluated temporal pixel masking on one UVC camera link. Across 8 common-setting geometries, the same 288 matched units per profile (12 content items $\times$ 8 geometries $\times$ 3 repeats) yield 94.5% mean recovery unprotected, 17.8% for readability-priority, and 5.6% for high-suppression. Paired content-cluster contrasts are 76.7 percentage points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]). Readability-priority retains 17.7% matched field-micro recovery and 95.0% P99 character recovery, while high-suppression meets its descriptive $< 10\%$ target.

Attack-oriented evaluation recovers substantially more information: a 150-frame temporal mean reaches 71.1% character recovery for readability-priority and 47.9% for high-suppression, rising to 79.7% and 53.9% at the common settings; in the full 9-geometry sensitivity pool, long-exposure recovery is 60.9% for readability-priority versus 47.3% unprotected. On-axis VLM probes also recover considerable large-font content, while the three-engine preprocessing oracle reaches 40.2% for readability-priority and 13.7% for high-suppression.

In a 56-participant user study, readability-priority masking retained 94.83% typing accuracy while reducing speed by 1.14 WPM; subjective readability, stability, and comfort averaged 3.79, 2.98, and 3.39 out of 5, confirming measurable usability costs concentrated in perceived stability for the readability-priority operating point. Detection and tracking stress tests show the same directional OCR-reduction pattern across four detectors. Future work should extend capture to additional cameras and displays, map VLM-aware adaptive profiles, and evaluate comfort over longer sessions.

DATA AVAILABILITY

The data and code that support the findings of this study are available from the corresponding author upon reasonable request.

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[PLACEHOLDER: Acknowledgment to be filled.]

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